# Nikos Litharis

AI Engineer — LLM Systems, RAG & Evaluation

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## SUMMARY

AI Engineer building production LLM systems in Python on Azure — RAG, multi-agent orchestration and evaluation pipelines — for public-sector clients including Greece's national power transmission operator (IPTO) and the Independent Authority for Public Revenue (AADE), building TAXIS, a legal-research assistant on the Microsoft Agent Framework.

## WORK EXPERIENCE

**Netcompany** Nov 2025 – Present

*AI Engineer*

- Owned the evaluation and UAT pipeline for TAXIS, AADE's production legal-research assistant, end to end: versioned scenario datasets covering adversarial, citation, temporal and incomplete-context challenge sets, with gold sources, a CLI runner that scores answers and citations, and evidence-promotion tooling with runtime provenance.
- Designed the evaluation suite behind retrieval improvements in IPTO's RAG knowledge-base assistant: 50 bilingual (Greek/English) cases over IEC standards documentation, used to iterate on prompts, chunking and embeddings.
- Architected a durable, resumable batch-ingestion pipeline for large legal corpora (OpenAI Batch API for embeddings and enrichment, Azure AI Search v3), with recovery for expired leases and concurrent admission, and a live dashboard for throughput and queue flow.
- Added voice interaction to IPTO's RAG assistant by integrating Azure Speech SDK STT/TTS with managed identity authentication, handling long-response streaming, timeouts and cancellation.
- Built Docker-containerized multi-agent (A2A) orchestration pipelines with LangChain, reused across delivery projects to cut manual coordination overhead.

**Military Service, Hellenic Army** Sep 2024 – Jun 2025

**The Prosperty** Feb 2024 – Mar 2024

*AI Development Intern*

- Cut listing-description production time from ~2 hours to under 30 seconds by building a GPT-3 generation tool in Python, adopted for new property listings.
- Raised generated-copy acceptance by fine-tuning language models on property data and iterating on training samples against tone and accuracy requirements.
- Integrated the tool into the live listing platform by shipping a REST API with the product and engineering teams.

## PROJECTS

**SkillPulse Agent**
Job-market analyser over Hacker News "Who is hiring?" threads: scheduled workers, LLM skill extraction into PostgreSQL, and a LangGraph agent answering only from SQL, with five read-only tools shared with an MCP server; scored by a 64-question benchmark with a grounding gate on every figure.

**MedLeaf RAG** ***(in progress)*** — `github.com/NickLitharis/medleaf-rag`
Bilingual (Greek/English) retrieval-augmented QA over official EU medicine information, with cited answers, medical safety guardrails, and a CI evaluation gate that blocks regressions.

**Anemos** — `github.com/NickLitharis/anemos`
Open-source operating system inspired by Windows XP, built as new code and validated for behavioural compatibility against XP itself: ABI, conformance and crypto gates, with DNS and DHCP codecs passing their suites.

**Haystack & Elasticsearch Q&A Search** — `github.com/NickLitharis/haystack-elastic`
Indexed scraped Q&A data into Elasticsearch and compared two retrieval pipelines — keyword (BM25) against semantic search with Haystack's EmbeddingRetriever, returning the top-5 answers by embedding similarity.

## EDUCATION

**BSc in Computer Science, Ionian University, Corfu** 2019 – 2024
Thesis: Text Generation from Medical Images — CNN encoder with an attention-based RNN decoder in TensorFlow, generating captions for medical images.

## SKILLS

**GenAI / LLM:** RAG, Prompt Engineering, Embeddings, LLM Evaluation, Fine-Tuning, Multi-Agent Orchestration (A2A), Microsoft Agent Framework, LangChain, Azure OpenAI, GPT APIs, Azure AI Search, PostgreSQL Vector Search
**Programming:** Python (NumPy, Pandas, Scikit-Learn, Keras), C# / .NET, TypeScript / Angular, SQL (Oracle, PostgreSQL)
**ML Engineering:** Docker, Docker Compose, GitLab CI, Git, Azure, REST APIs, Automated Regression Testing (.NET, Angular, Python), LLM Evaluation Harnesses, Model Monitoring
**AI-Assisted Dev:** Claude Code, Pi (multi-model: GPT, DeepSeek), agentic coding workflows